

Problem 7

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Problem 1 Find the average value of the function $g(t) = 4te^{-t^2}$ on the interval $[0, 3]$.

Explanation. We know $g_{\text{avg}} = \frac{1}{b-a} \int_a^b g(t) dt$ with $[a, b]$ as $[0, 3]$. We'll make a substitution, $w = t^2$ with $dw = 2t dt$. Notice that when $t = 0$, $w = (0)^2 = 0$ and when $t = 3$, $w = (3)^2 = 9$.

$$\begin{aligned}
 g_{\text{avg}} &= \frac{1}{3-0} \int_0^3 4te^{-t^2} dt \\
 &= \frac{1}{3} \int_0^9 2e^{-w} dw \\
 &= \frac{1}{3} \left[-2e^{-w} \right]_0^9 \\
 &= \frac{1}{3} [(-2e^{-9}) - (-2e^0)] \\
 &= \frac{1}{3} \left(2 - \frac{2}{e^9} \right)
 \end{aligned}$$